

AN ARTIFICIAL NEURAL NETWORK REPRESENTATION OF THE SABR STOCHASTIC VOLATILITY MODEL

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MOTIVATION

- ▶ The stochastic-volatility SABR model introduced by Hagan et al. (2002) is one of the most widely used models for fitting and extrapolating implied-volatility smiles.
- ▶ The Hagan closed-form approximation is extremely fast but may lose accuracy in:
 - Long maturities,
 - Extreme strikes,
 - High vol-of-vol environments,
- ▶ Numerical PDE solvers (2FDM) provide high accuracy but at prohibitive computational cost.
- ▶ **Objective:** Build neural surrogates that are:
 - **As fast** as the Hagan formula,
 - **As accurate** as a finite-difference solver,
 - **Robust** across the full elasticity range $\beta \in [0, 1]$.

ORIGINAL APPROACH (McGHEE, 2020)

► Two-stage surrogate construction:

1. Train on Hagan IVs (fast, analytical baseline).
2. Fine-tuning on 2FDM implied volatilities (high-fidelity numerical benchmark).

► Benefits:

- Produces a globally smooth implied-volatility surface.
- Significantly improves accuracy over Hagan.
- Achieves large computational speed-ups relative to PDE solvers.

OUR CONTRIBUTIONS

- ▶ Replicated Phase 1 training for:
 - $\beta = 1$ (lognormal case),
 - Full elasticity range $\beta \in [0, 1]$,
 - Width ablation study: quantified the link between accuracy and network capacity.
- ▶ Extended Phase 2 fine-tuning to:
 - $\beta \in [0, 1]$,
 - Comparison of 2FDM vs. conditional-integration datasets.

THEORY BACKGROUND — THE SABR MODEL

The SABR model is defined as follows:

$$dF_t = \sigma_t F_t^\beta dW_t, \quad d\sigma_t = \xi \sigma_t dZ_t, \quad d\langle W, Z \rangle_t = \rho.$$

► **Elasticity parameter β :** controls how volatility reacts to changes in the forward.

- $\beta = 0$ — Normal model,
- $\beta = 1$ — Lognormal model,
- $0 < \beta < 1$ — CEV-like intermediate regime.

► **Impact on smile:**

- Lower $\beta \rightarrow$ steeper skew and heavier left tail,
- Higher $\beta \rightarrow$ smoother, more symmetric smiles,
- Overall shape governed jointly by (β, ξ, ρ) .

THEORY BACKGROUND — HAGAN IMPLIED VOLATILITY

For a European option with forward F_0 , strike K , and maturity T , the Hagan et al. (2002) approximation is:

$$\sigma_{\text{imp}}(K, F_0, T) = \frac{\alpha z}{q(K, F_0) x(z)} \left[1 + \left(\frac{(1-\beta)^2}{24} \frac{\alpha^2}{(F_0 K)^{1-\beta}} + \frac{\rho \beta \xi \alpha}{4(F_0 K)^{(1-\beta)/2}} + \frac{2-3\rho^2}{24} \xi^2 \right) T + \dots \right].$$

$$z = \frac{\xi}{\alpha} \frac{F_0^{1-\beta} - K^{1-\beta}}{1-\beta}, \quad x(z) = \log \left(\frac{\sqrt{1-2\rho z + z^2} + z - \rho}{1-\rho} \right).$$

- ▶ Very **fast** to evaluate,
- ▶ Accuracy **degrades** for long maturities, extreme strikes, or large vol-of-vol.

THEORY BACKGROUND — NEURAL NETWORK ARCHITECTURE

- ▶ Fully-connected feed-forward network predicting IVs at 10 strikes,

- ▶ Input features:

$$(T, \sigma_0, \xi, \rho, \beta, x_1, \dots, x_{10}),$$

- ▶ Smooth and differentiable mapping,

- ▶ The Universal Approximation Theorem ensures representational power.

THEORY BACKGROUND — SABR PDE AND 2FDM

Under the forward measure:

$$V_t = A(V) + B(V) + C(V),$$

$$A(V) = \frac{1}{2}\sigma^2 F^{2\beta} V_{FF}, \quad B(V) = \frac{1}{2}\xi^2 \sigma^2 V_{\sigma\sigma}, \quad C(V) = \rho\xi\sigma^2 F^\beta V_{F\sigma}.$$

- ▶ Solve on a 2D grid in (F, σ) ,
- ▶ Craig–Sneyd ADI scheme ($\theta = 1/2$):
 - Alternating implicit sweeps in F and σ ,
 - Explicit correction for the mixed term C ,
 - Restores second-order accuracy.
- ▶ Produces a smooth, coherent surface used as a high-fidelity training target.

THEORY BACKGROUND — CONDITIONAL INTEGRATION

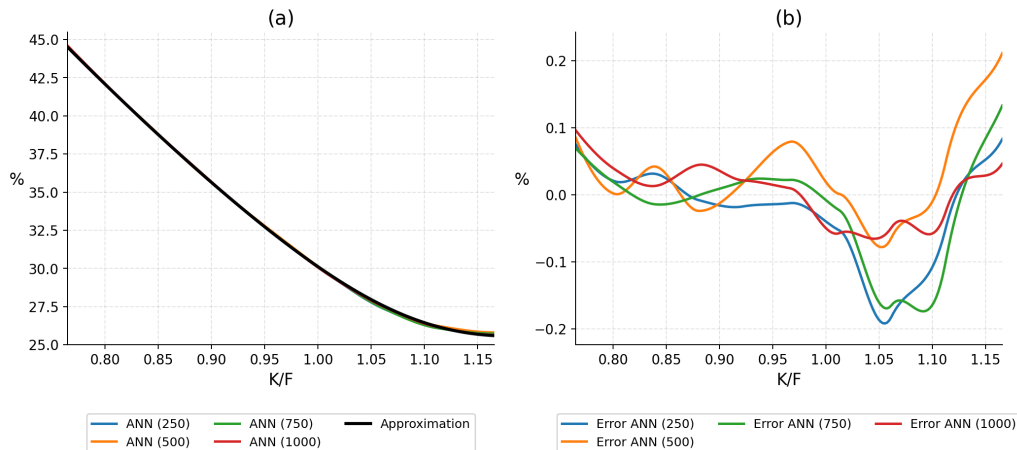
The volatility is lognormal for any β :

$$d\sigma_t = \xi \sigma_t dW_t^\sigma \quad \Rightarrow \quad \log \sigma_T \sim \mathcal{N}(\log \sigma_0 - \frac{1}{2}\xi^2 T, \xi^2 T).$$

- ▶ Nested quadrature over $(\sigma_T, \vartheta(T))$, where $\vartheta(T) = \frac{1}{T} \int_0^T \sigma(t)^2 dt$,
- ▶ Accurate pointwise, but the resulting IV surface lacks global smoothness,
- ▶ Leads to unstable neural refinement.

RESULTS — PHASE 1 (HAGAN TRAINING)

ANN smiles vs SABR approximation ($T = 14D$, $\sigma_0 = 30\%$, $\xi = 160\%$, $\rho = -60\%$)



Short-maturity regime ($T = 14$ days, $\sigma_0 = 30\%$, $\xi = 160\%$, $\rho = -60\%$).

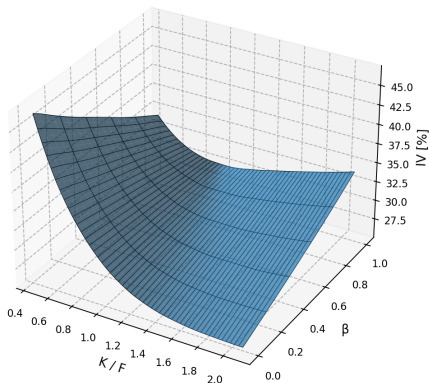
Left: ANN-predicted volatility smiles vs. Hagan approximation. Right: corresponding percentage errors across strikes.

- Very accurate reproduction of the Hagan implied volatilities.

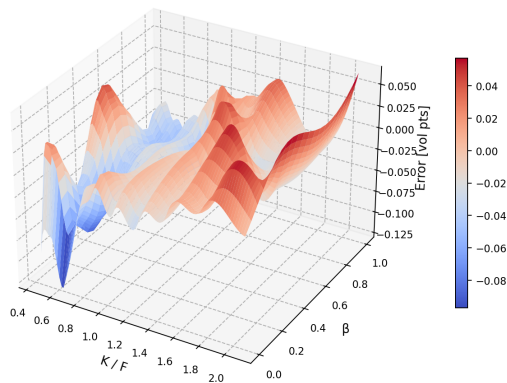
RESULTS — PHASE 1 (HAGAN TRAINING FOR $\beta \in [0, 1]$)

3D smile vs β (fig 3)

SABR vs ANN IV surface
($T = 6M$, $\sigma_0 = 30\%$, $\xi = 40\%$, $\rho = 0\%$)



ANN - SABR (vol points)



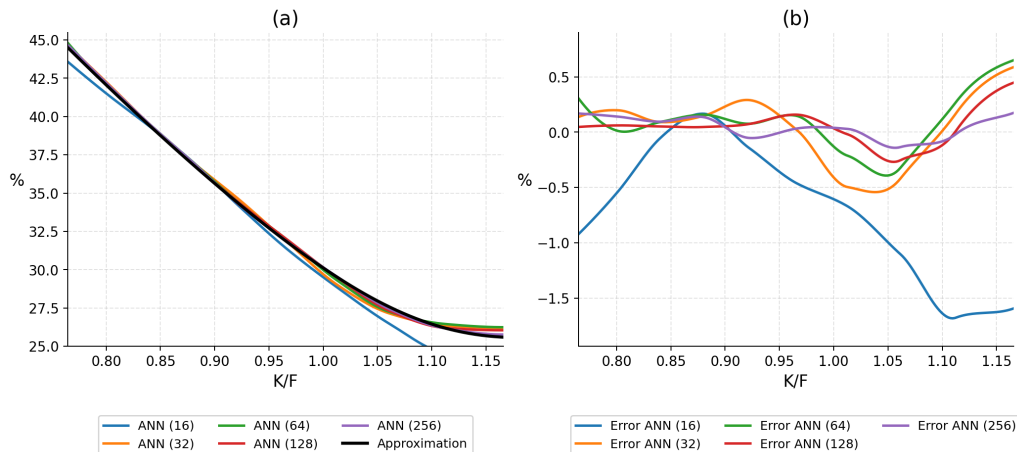
Generalized elasticity case ($\beta \in [0, 1]$), network width $W = 1000$.

Left: ANN-predicted implied-volatility surface vs. Hagan formula. Right: corresponding pointwise percentage errors across $(K/F, \beta)$.

- Consistent behaviour for $\beta = 1$ and $\beta \in [0, 1]$.

RESULTS — PHASE 1 (HAGAN TRAINING - WIDTH ABLATION)

ANN smiles vs SABR approximation ($T = 14D$, $\sigma_0 = 30\%$, $\xi = 160\%$, $\rho = -60\%$)



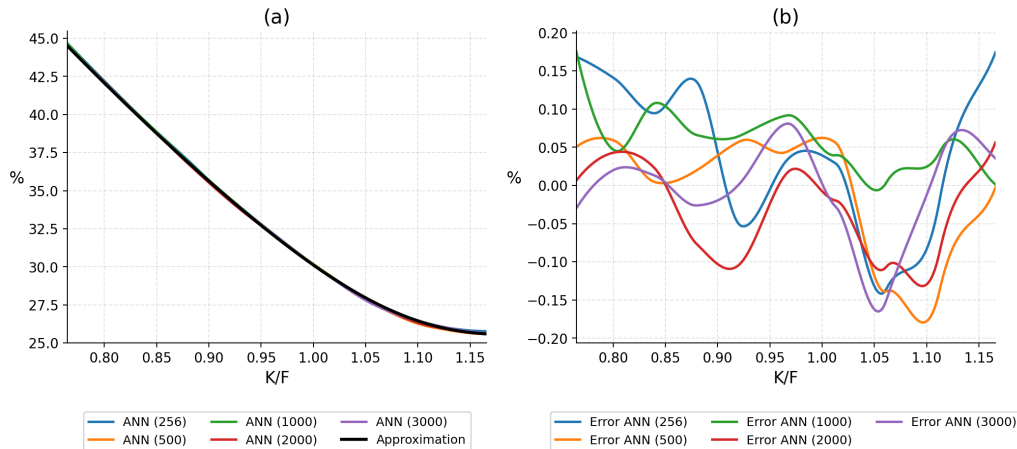
Short-maturity regime: ANN smiles and errors for small-medium hidden-layer widths.

Left: ANN-predicted volatility smiles vs. Hagan approximation. Right: corresponding percentage errors across strikes.

► Small networks underfit,

RESULTS — PHASE 1 (HAGAN TRAINING - WIDTH ABLATION)

ANN smiles vs SABR approximation ($T = 14D$, $\sigma_0 = 30\%$, $\xi = 160\%$, $\rho = -60\%$)

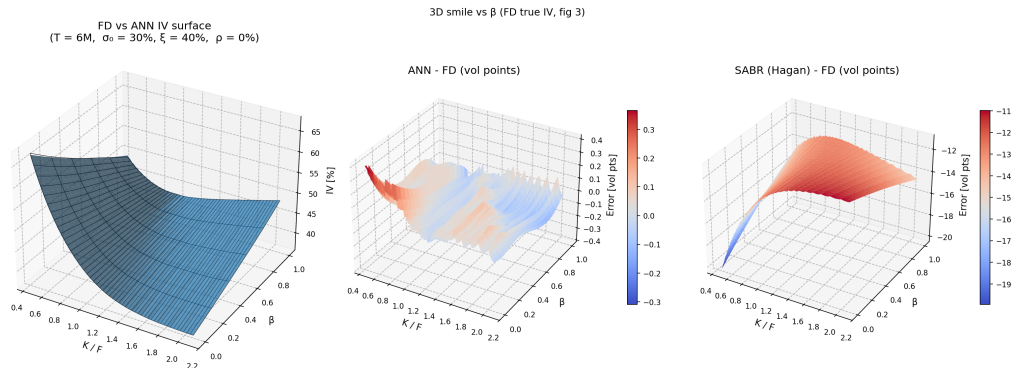


Short-maturity regime: ANN smiles and errors for medium–large hidden-layer widths.

Left: ANN-predicted volatility smiles vs. Hagan approximation. Right: corresponding percentage errors across strikes.

- Accuracy stabilises around $W \approx 500$,
- Little gain beyond $W > 1000$.

RESULTS — PHASE 2 (2FDM FINE-TUNING)



Generalized elasticity case ($\beta \in [0, 1]$), FD training target, network width $W = 250$.

Left: finite-difference implied-volatility surface. Centre: ANN–FD pointwise errors. Right: Hagan–FD pointwise errors.

- ▶ 2FDM produces a smooth, coherent, PDE-consistent surface,
- ▶ Sub-basis-point precision reached for $W \geq 500$,
- ▶ Matches behaviour reported in McGhee (2020).

NUMERICAL ISSUE & ADI CORRECTION

- ▶ We solve the SABR PDE in the form

$$\partial_t V = (A + B + C)V, \quad AV = \frac{1}{2}\sigma^2 F^{2\beta} V_{FF}, \quad BV = \frac{1}{2}\xi^2 \sigma^2 V_{\sigma\sigma}, \quad CV = \rho\xi\sigma^2 F^\beta V_{F\sigma}.$$

- ▶ Our **first FD implementation** used a simplified Douglas–type ADI step:

- Predictor with C treated *explicitly*,
- One implicit sweep in F for A ,
- One implicit sweep in σ for B .

- ▶ A Taylor expansion of this scheme shows that one time step is

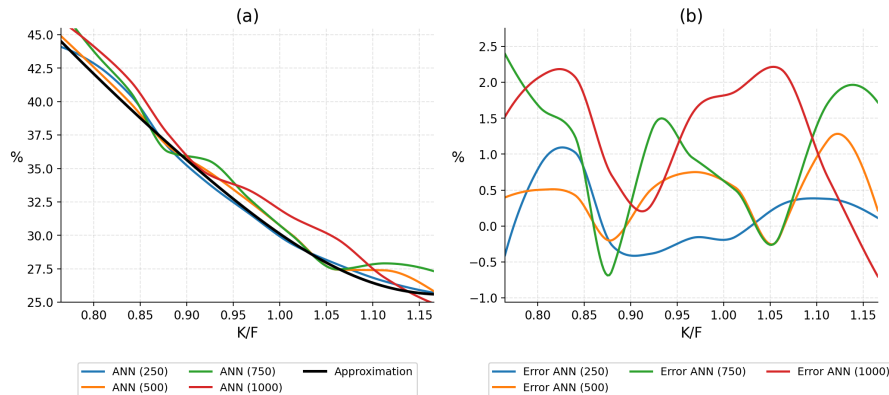
$$V^{n+1} = V^n + \Delta t(A + B + C)V^n + \Delta t^2(\alpha_A A^2 + \alpha_B B^2 + \alpha_{AB}AB + \dots)V^n,$$

i.e. the pure diffusion operators A and B are effectively applied *more than once* per step.

- ▶ Since A and B are second–derivative operators, these extra $\mathcal{O}(\Delta t^2)$ terms act as **artificial diffusion**: the surface is over-smoothed and call prices are systematically too high $\Rightarrow$ implied vols are shifted upwards by several vol points.
- ▶ We therefore replaced this with a **Craig–Sneyd ADI** scheme ($\theta = 1/2$) that:
 - Adds a dedicated correction step for the mixed term C ,
 - Restores second-order accuracy in time,
 - Removes the artificial diffusion bias in the FD implied vols.

RESULTS – PHASE 2 (CONDITIONAL-INTEGRATION FINE-TUNING)

ANN smiles vs SABR approximation ($T = 14D$, $\sigma_0 = 30\%$, $\xi = 160\%$, $\rho = -60\%$)



Short-maturity regime ($T = 14$ days, $\sigma_0 = 30\%$, $\xi = 160\%$, $\rho = -60\%$).

Left: ANN-predicted volatility smiles vs. integration-based values. Right: corresponding percentage errors across strikes.

- ▶ Computationally efficient and pointwise accurate,
- ▶ But IV surface lacks global coherence,
- ▶ Not suitable as refinement training target.

CONCLUSION

- ▶ Warming up training (Hagan) is robust across the full β -range,
- ▶ Width sweep identifies minimal architecture ≈ 500 ,
- ▶ 2FDM fine-tuning yields highly accurate, smooth surfaces,
- ▶ Conditional integration underperforms as training target.
- ▶ Corrected FD solver (Craig–Sneyd) improves numerical fidelity.